

SWIS

SELF - WORKING
INSTALLABLE ANY WHERE ON EARTH
SUSTAINABLE

WHEN SURVIVAL BECOMES HOME A STORY

YEAR 3226

CENTURIES OF CLIMATE CHANGE HAVE TRANSFORMED EARTH. AS GLOBAL TEMPERATURES RISE, ANTARCTICA HAS LOST MUCH OF ITS ICE, LEAVING BEHIND ONLY A FRAGILE CORE.

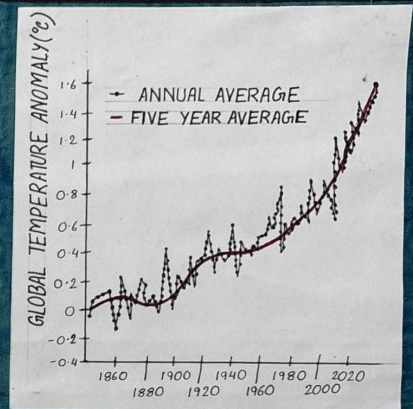
DESPITE THESE CHANGES, ANTARCTICA REMAINS THE GLOBAL HUB FOR SCIENTIFIC RESEARCH AND PEACE. TO PROTECT ITS REMAINING LAND WHILE SUPPORTING THE SCIENTISTS WHO CONTINUE THEIR WORK, A NEW IDEA TAKES SHAPE - A RING OF "SELF-SUSTAINING

HOUSE ISLANDS" SURROUNDING THE CONTINENT.

THESE FLOATING HOMES PROVIDE A SAFE HABITAT FOR SCIENTISTS AND THEIR FAMILIES, FORMING A PROTECTIVE COMMUNITY AROUND ANTARCTICA RATHER THAN OCCUPYING ITS REMAINING LAND.

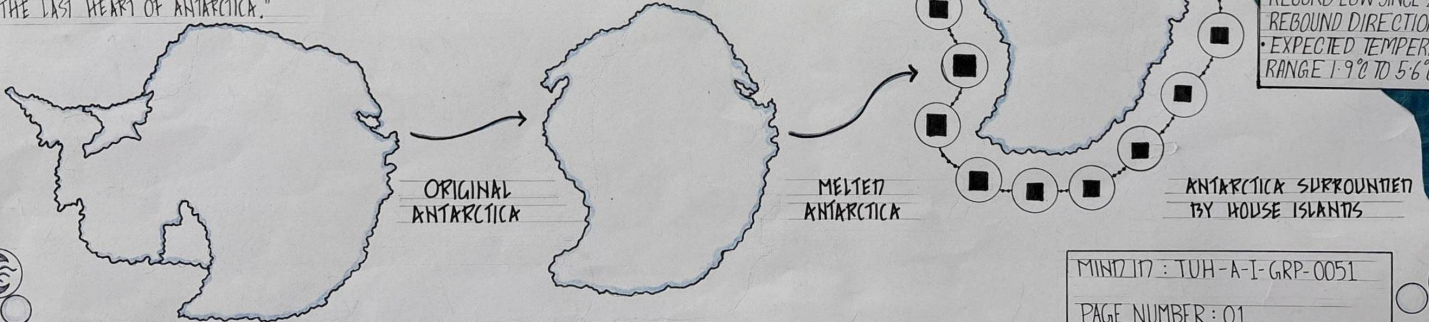
"THIS IS MORE THAN A HOUSE. IT IS A HOME DESIGNED TO PRESERVE THE LAST HEART OF ANTARCTICA."

FEATURE	RUSSIA	EUROPE	USA	ANTARCTICA
PERMANENT POPULATION	144 MILLION	740 MILLION	340 MILLION	0
TYPICAL WINTER	-10°C to -40°C	-5°C to +10°C	-30°C to -20°C	-40°C to -80°C
SUMMER	15°C - 30°C	15°C - 35°C	20°C - 45°C	-20°C - 5°C
FARMING	EXTENSIVE	EXTENSIVE	EXTENSIVE	NO OUTDOORS
PERMANENT RESIDENTS	YES	YES	YES	NO



- SOURCES:
- NASA SCIENCE
- NOAA CLIMATE GOVT.
- IPCC

- 0.1% OF ICE HAS MELTED OVER YEARS.
- ANTARCTICA LOSES 100 - 135 BILLION TONS OF ICE / YEAR.
- RECORD LOW SINCE 2023, REBOUND DIRECTION.
- EXPECTED TEMPERATURE RANGE 1.9°C TO 5.6°C



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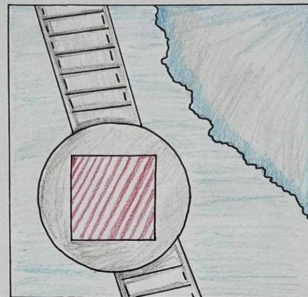
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CONCEPT

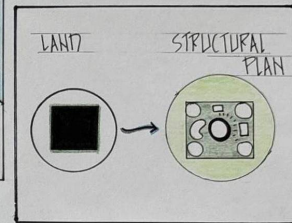
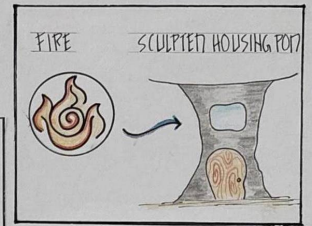
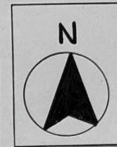
THE DESIGN IS INSPIRED BY THE FIVE ELEMENTS OF NATURE - EARTH, WATER, AIR, FIRE AND SPACE. EACH ELEMENT INFLUENCES A UNIQUE PART OF THE ARCHITECTURE, CREATING A SUSTAINABLE AND RESILIENT HABITAT FOR THE FUTURE OF ANTARCTICA.

- **EARTH** - STABILITY AND STRENGTH, SQUARE PLAN WITHIN A CIRCULAR FORM
- **WATER** - FLOW, ADAPTABILITY & RESILIENCE, STONE FORM CARVED BY SKILLED CRAFTSMEN
- **AIR** - SYMBOLIZES OPENNESS AND SURROUNDS EVERYTHING. MOON ACTS AS THE LANDMARK ELEMENT OF AIR AND THE DESIGN.
- **FIRE** - WARMTH AND HUMAN CONNECTION, COMFORTABLE RESIDENTIAL LIVING PORTS IN EXTREME CLIMATE.
- **SPACE (AKĀŚĀ)** - SYMBOLIZES THE ESSENCE OF ARCHITECTURE, CONNECTS ALL THE ELEMENTS THROUGH THOUGHTFULLY DESIGNED SPACES.

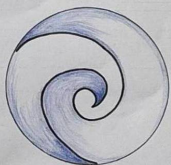
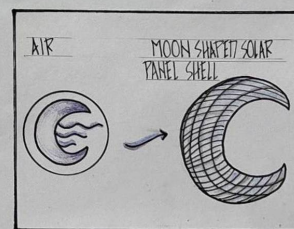
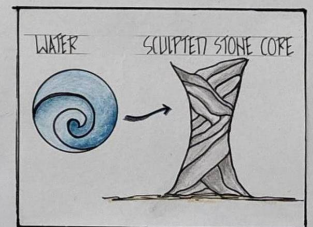
KEY PLAN



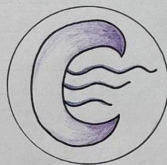
(NTS)



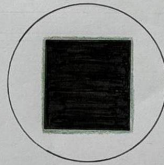
FORM DEVELOPMENT



WATER



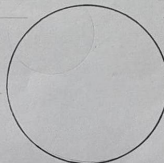
AIR



LAND



FIRE



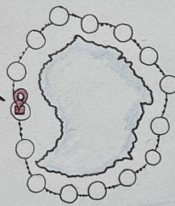
SPACE

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SITE ANALYSIS

- SITE: COASTAL ISLAND, ANTARCTICA
 SITE LOCATION
- CONTINENT - ANTARCTICA
 - REGION - COASTAL ICE ISLAND
 - LATITUDE - APPROXIMATELY 70°S
 - SURROUNDINGS - SOUTHERN OCEAN
 PERMANENT ICE SHEET
 GLACIERS
 ROCKY OUTCROPS



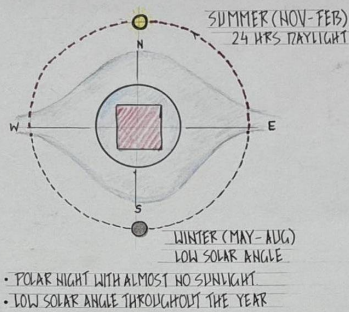
CLIMATE ANALYSIS

SUMMER -5°C to 2°C	WINTER -35°C to -60°C	SNOWFALL THROUGHOUT THE YEAR
HUMIDITY LOW	RAINFALL VERY LOW (POLAR DESERT)	WIND STRONG WIND

CHALLENGES

- EXTREME COLT
- STRONG WINDS
- MONTHS OF DARKNESS
- MOVING ICE
- NO INFRASTRUCTURE

SUNPATH ANALYSIS

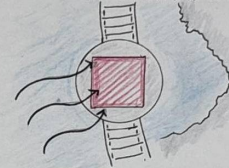


TOPOGRAPHY

- MOSTLY FLAT ICE SURFACE
 - GENTLE SLOPE
 - ROCKY PATCHES SUITABLE FOR FOUNDATIONS
 - THICK ICE LAYER
- GENTLE SLOPE TOWARDS THE SEA

VEGETATION		
MOSS	LICHEN	ALGAE
WILDLIFE		
PENGUIN	SEAL	WHALE
PEBBLES	SKUAS	MICROBACTERIA

WIND ANALYSIS



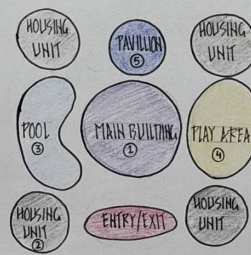
- PREVAILING WIND: KATABATIC WINDS
 DIRECTION: INTERIOR ICE SHEET → COAST
 SPEED: 80-150 KM/H

DESIGN RESPONSE

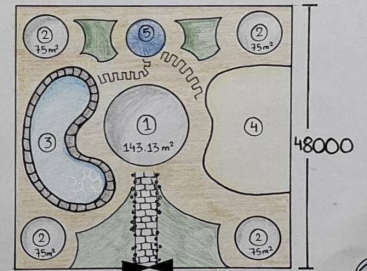
- CURVED BUILDING FORM
- WIND BARRIERS
- ENTRANCE AWAY FROM PREVAILING WIND

OPPORTUNITIES

MAJN SOLAR GAIN	WIND PROTECTION	ELEVATED FORM
SELF-SUSTAINING	STEAL FOR RESEARCH	NATURAL WATER SOURCE



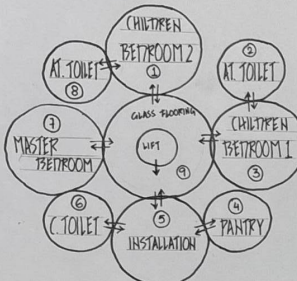
SITE ZONING
 NOT TO SCALE (NTS)



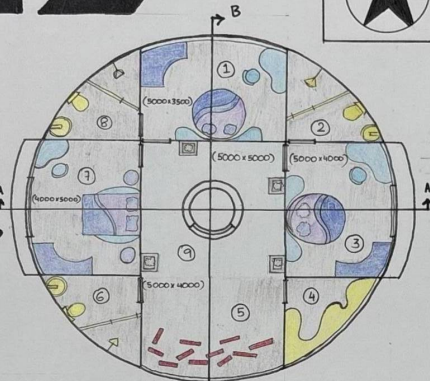
SITE PLANNING
 SCALE = 1:500

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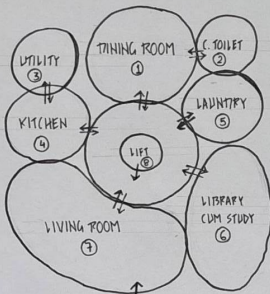
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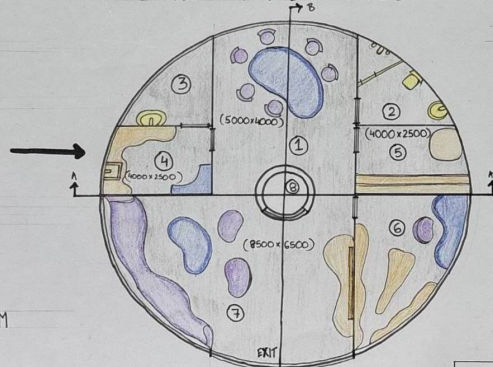
1ST FLOOR BUBBLE DIAGRAM



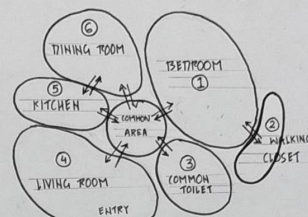
1ST FLOOR PLAN [SCALE-1:100]



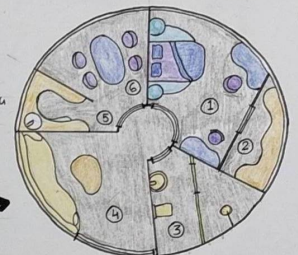
GROUND FLOOR BUBBLE DIAGRAM



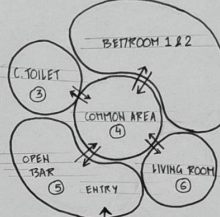
GROUND FLOOR PLAN [SCALE-1:100]



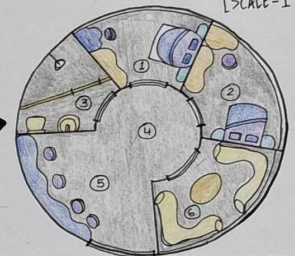
SERVANTS & GRAND PARENT'S HOUSE BUBBLE DIAGRAM



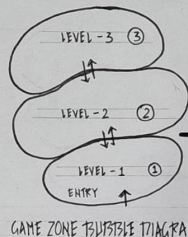
SERVANTS & GRAND PARENT'S HOUSE PLAN [SCALE-1:100]



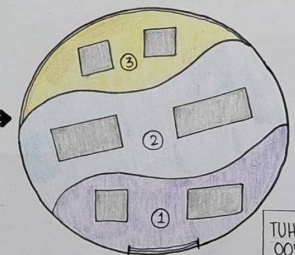
GUEST HOUSE BUBBLE DIAGRAM



GUEST HOUSE PLAN [SCALE-1:100]



GAME ZONE BUBBLE DIAGRAM

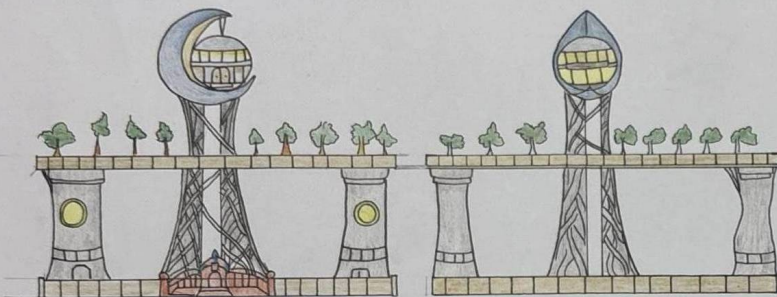


[SCALE-1:100] GAME ZONE PLAN

ALL DIMENSIONS ARE IN MM

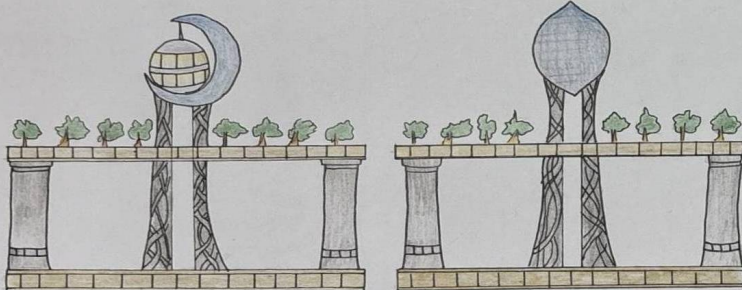
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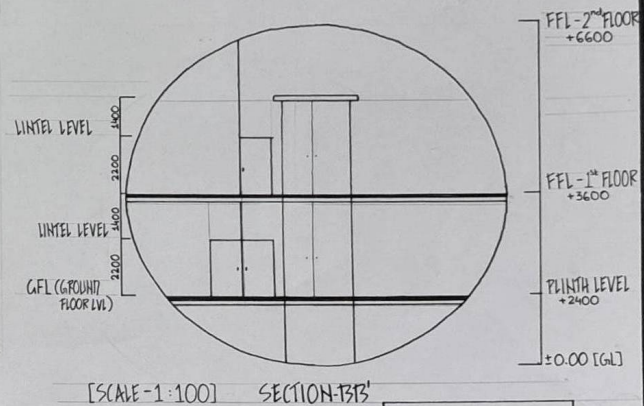
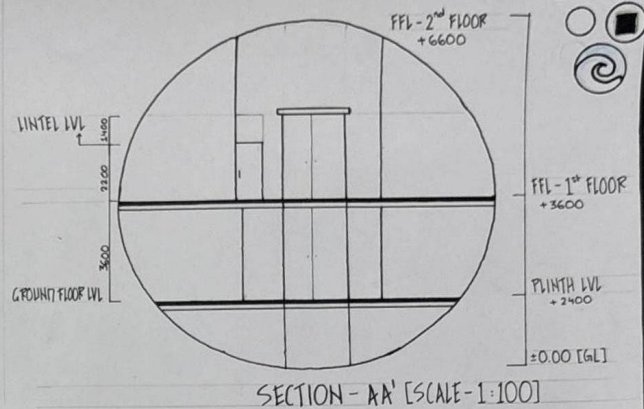
FRONT ELEVATION [SCALE-1:500]

SIDE ELEVATION 1 [SCALE-1:500]



[SCALE-1:500] REAR ELEVATION -2

SIDE ELEVATION -2 [SCALE-1:500]



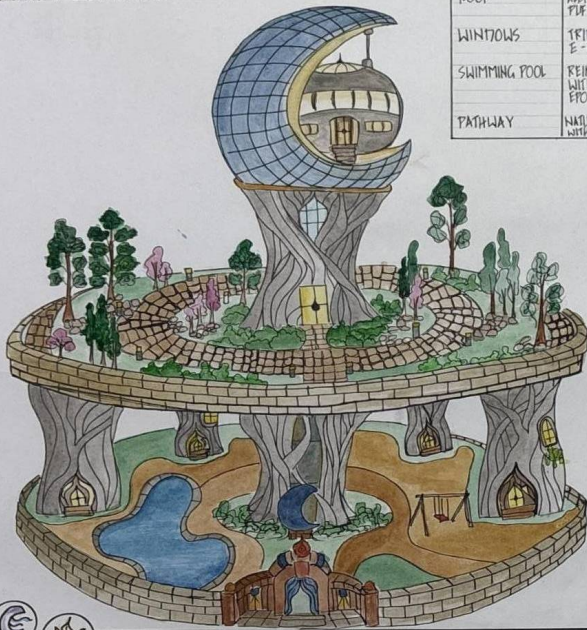
ALL DIMENSIONS ARE IN MM

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3D VIEW

VANCE'S RESIDENCE



MATERIAL CHART

ELEMENT	MATERIAL
MAIN HOUSE WALLS	INSULATED SANDWICH PANELS WITH THICKER CLADDING
STRUCTURE	STRUCTURAL STEEL FRAME
ROOF	ALUMINUM ROOFING WITH PTFE INSULATION
WINDOWS	TRIPLE-GLAZED LOW-E GLASS
SWIMMING POOL	REINFORCED CONCRETE WITH WATERPROOF EPOXY FINISH
PATHWAY	NATURAL STONE PAVES (GRABED) WITH ANTI-SLIP FINISH

KEY DETAILS



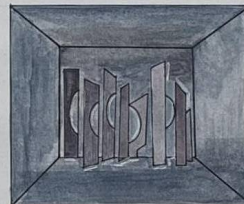
MOON-THEMED GAZEBOS
A QUIET SPACE FOR
STARGAZING, MEDITATION
AND FAMILY GATHERINGS.



LIFT CORE
DIRECT LIFT FROM
ENTRANCE TO THE
MAIN HOUSE



PLAY AREA
FOR FAMILY
RECREATION AND
ACTIVE LIVING.



FEEL OF THE MOON
AN ABSTRACT INSTALLATION THAT CAPTURES
THE MOON'S ESSENCE THROUGH LIGHT, SHADOW,
REFLECTION AND STILLNESS.

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